Spatial Data

By

Dr. Meenakshi Devi

EES Department, IISER Mohali

1. Introduction to Spatial Data

•**Spatial data** (also called geospatial data) refers to information about the **location, shape, and relationships of geographic features**.

•Essential to **mapping, spatial analysis**, and **geographic modeling** in GIS.

Key Components:

- **Location**: Coordinates (latitude, longitude)
- **Attributes**: Descriptive information (e.g., population, land use)
- **Topology**: Spatial relationships (adjacency, connectivity)

2. Types of Spatial Data in GIS

A. Vector Data

- Represents **discrete features** using geometric shapes.
- Built using **points, lines, and polygons**.

a) Point Data

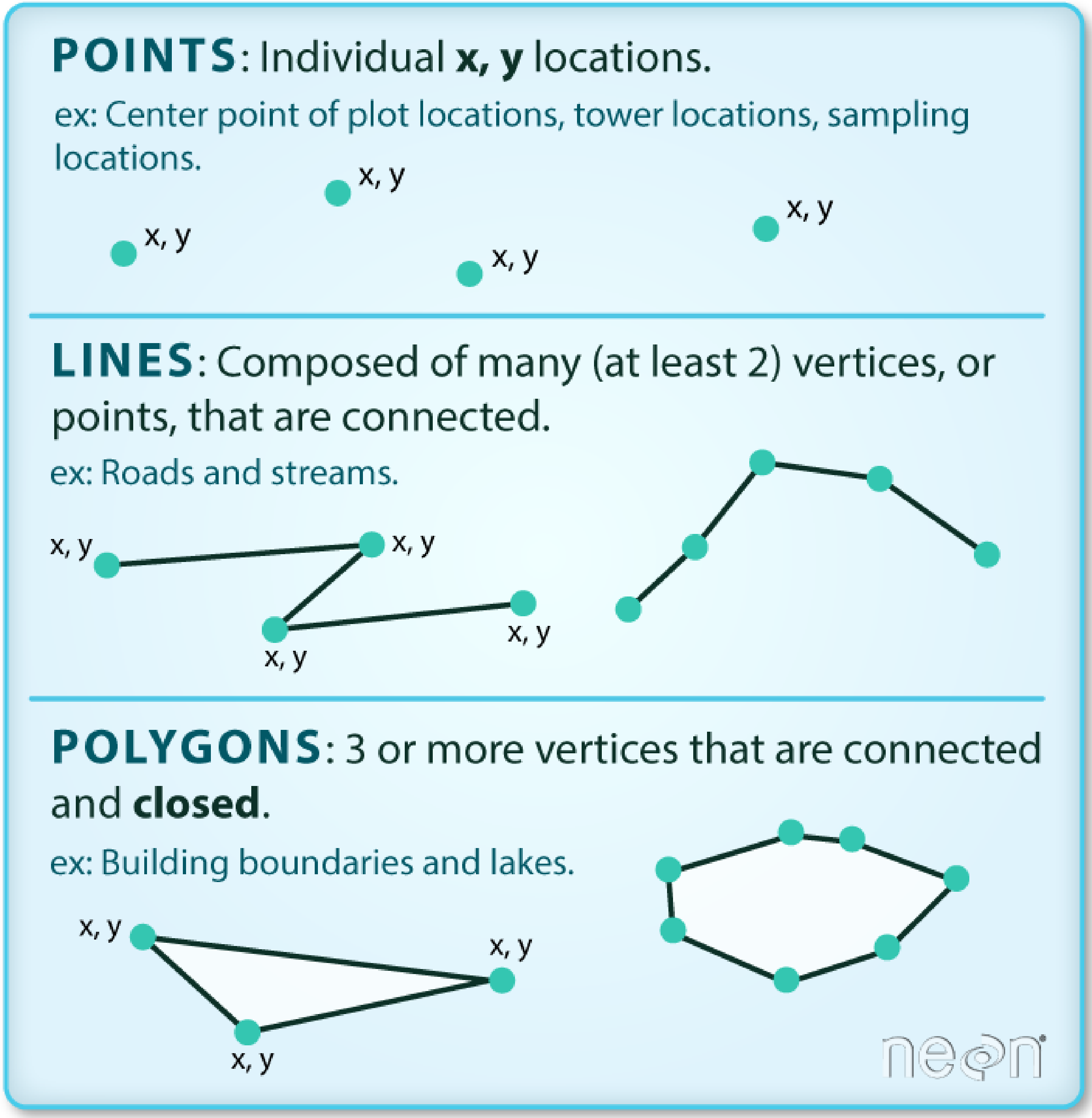
- Represents a **single location** (0-dimensional).
- Examples: Wells, schools, trees, GPS coordinates

b) Line Data

- Represents **linear features** (1-dimensional).
- Has length but no area.
- Examples: Rivers, roads, pipelines

c) Polygon Data

- Represents **area features** (2-dimensional).
- Encloses space.
- Examples: Lakes, parks, land parcels, administrative boundaries

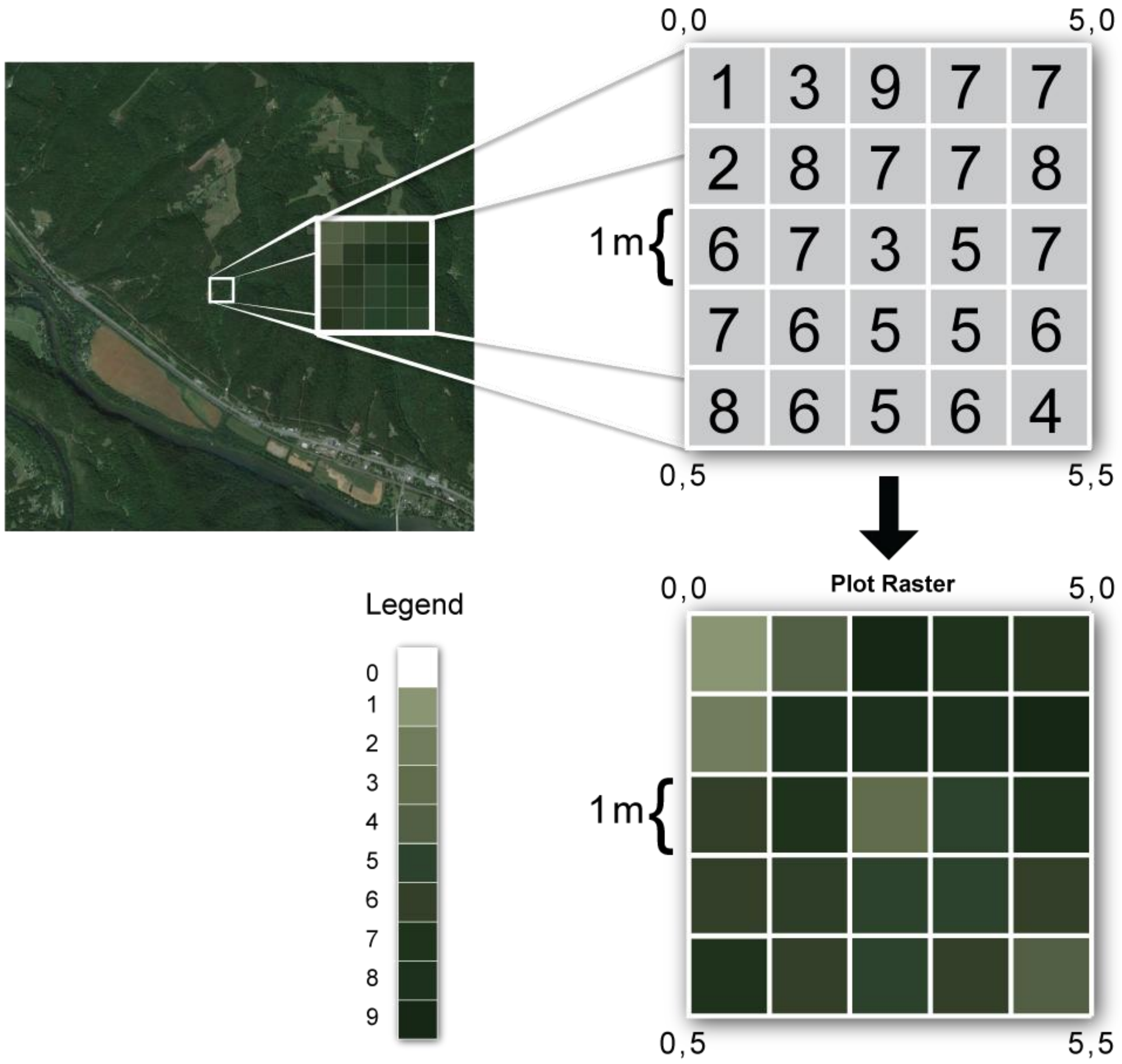


B. Raster Data

- Represents the world as a **grid of cells** (pixels).
- Each cell has a value representing **an attribute** (e.g., elevation, temperature, land cover).
- Used for **continuous data**.

Raster Characteristics:

- Cells arranged in rows and columns
- Resolution = cell size (smaller cells → higher detail)
- Examples: Satellite images, DEM (Digital Elevation Model), land cover maps



Comparison: Vector vs Raster Data

Feature	Vector Data	Raster Data
Structure	Points, lines, polygons	Grid of cells (pixels)
Data type	Discrete features	Continuous features
Storage efficiency	Compact for discrete data	May be large for high-resolution data
Precision	High positional accuracy	Depends on resolution
Analysis type	Network, overlay, proximity	Surface, terrain, image analysis
Example use	Urban planning, cadastral maps	Environmental modeling, remote sensing

Attribute Data

- Stored in **attribute tables** linked to spatial features.
- Each feature has a unique ID (FID or Object ID).
- Describes characteristics like:
  - Population, land use type, soil type, elevation, etc.

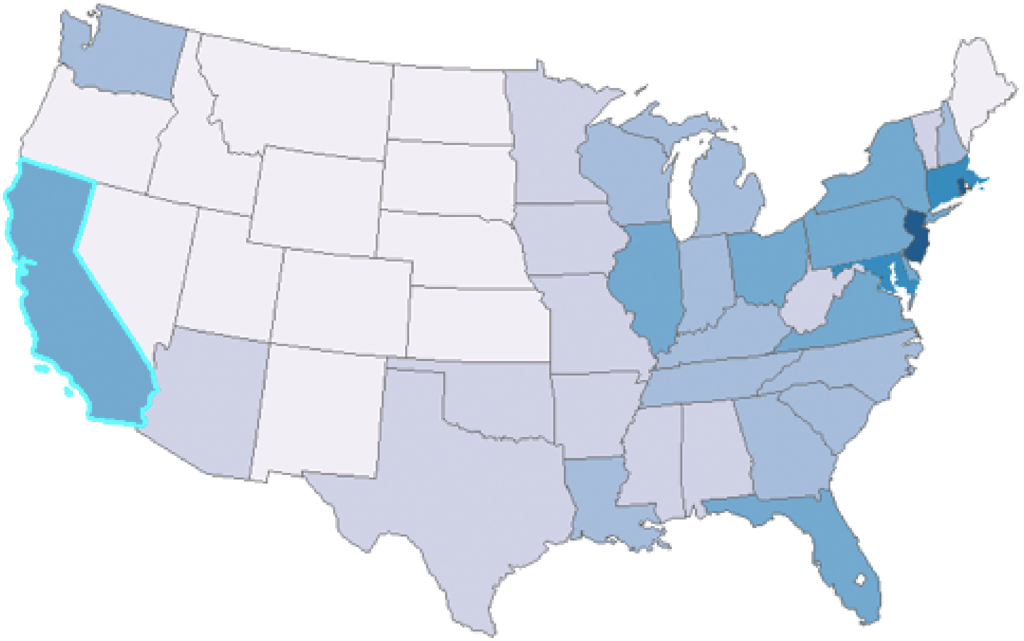
Table

Population Density

FID	Shape	STATE	NAME	FIPS	LON	LAT
0	Polygon	AK	Alaska	02	-152.24099	64.24018
1	Polygon	AL	Alabama	01	-86.82675	32.79353
2	Polygon	AR	Arkansas	05	-92.4392	34.89977
3	Polygon	AZ	Arizona	04	-111.66457	34.29323
4	Polygon	CA	California	06	-119.60818	37.24537
5	Polygon	CO	Colorado	08	-105.54783	38.99855
6	Polygon	CT	Connecticut	09	-72.72623	41.62196
7	Polygon	DC	District of Columbia	11	-77.01464	38.90932
8	Polygon	DE	Delaware	10	-75.50592	38.99559
9	Polygon	FL	Florida	12	-82.50941	28.67437
10	Polygon	GA	Georgia	13	-83.44848	32.65155
11	Polygon	HI	Hawaii	15	-156.34744	20.24924
12	Polygon	IA	Iowa	19	-93.50003	42.07463
13	Polygon	ID	Idaho	16	-114.65933	44.38905
14	Polygon	IL	Illinois	17	-89.19838	40.06501
15	Polygon	IN	Indiana	18	-86.27548	39.90801
16	Polygon	KS	Kansas	20	-98.38019	38.48471

26 (1 out of 55 Selected)

Population Density



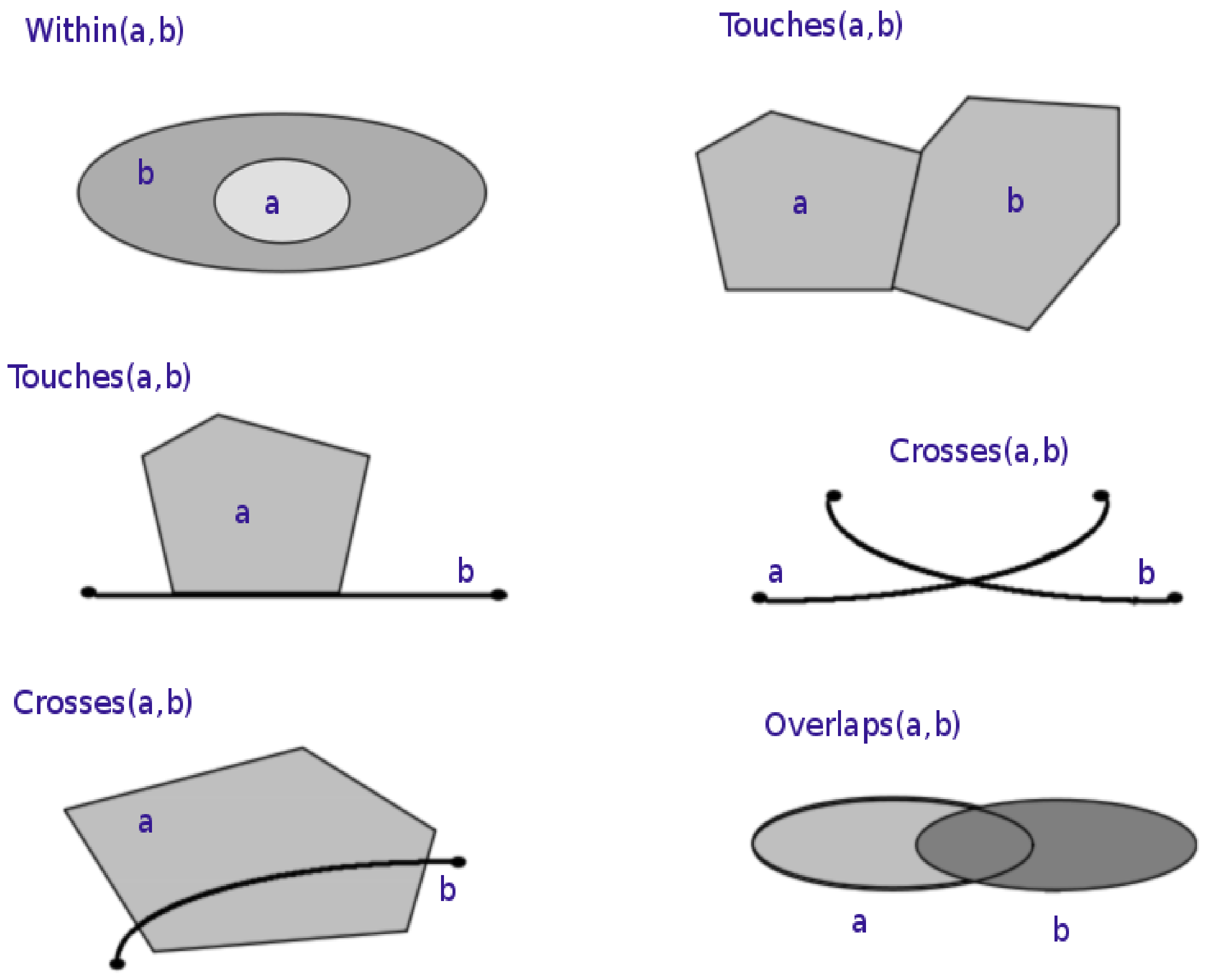
Topology in GIS

Definition:

- **Topology** in GIS refers to the spatial relationships between geometric features such as points, lines, and polygons, and how these features share geometry.

Importance:

- Ensures **data integrity** in spatial datasets.
- Supports **spatial analysis** and **network modeling**.
- Essential for **editing**, **querying**, and **validating** vector data.



Topological Concepts

Concept	Description
Connectivity	How features (e.g., roads, rivers) are connected (e.g., intersections, junctions).
Adjacency	When polygons share a common edge (e.g., neighboring land parcels).
Containment	One feature lies completely within another (e.g., a lake within a park).
Planar Enforcement	Ensures no gaps or overlaps between polygons in a dataset.

Vector Data Models and Topology

<u>Spaghetti Model</u> <u>(Non-topological)</u>

- Stores features as independent geometries.
- Easy to create but lacks spatial relationships.
- No enforcement of connectivity or adjacency.

<u>Topological model</u>

- Stores spatial relationships explicitly.
- Features share boundaries (e.g., adjacent polygons share edges).
- Used in systems like ArcInfo cov

Components:

- **Nodes**: Points where lines begin or end.
- **Arcs**: Directed lines connecting nodes.
- **Polygons**: Formed by arcs.
- **Topology tables**: Store relationships (e.g., arc-node, polygon-arc).

Topology Rules

Rule Type	Example
Must Not Overlap	Two land parcels cannot share the same space.
Must Not Have Gaps	No holes between adjacent land parcels.
Must Be Covered By Feature Class	All bus stops must lie on bus routes.
Must Not Self-Intersect	A road line feature should not loop or cross itself.
Must Be Inside	A building must lie within a parcel boundary.

3D and Temporal Spatial Data (Advanced Types)

a) 3D Spatial Data

- Adds **elevation or depth** to features (x, y, z).
- Used in geology, mining, architecture.

b) Temporal Data

- Adds a **time dimension** to track changes over time.
- Example: Land use change from 1990 to 2020

File Formats of Spatial Data

Vector Formats:

- Shapefile (.shp)
- GeoJSON
- KML (Keyhole Markup Language)
- GDB (Geodatabase)

Raster Formats:

- GeoTIFF
- IMG (ERDAS Imagine)
- GRID
- NetCD

Spatial Reference and Projection

- Spatial data must have **coordinate system** information:
  - Geographic: Latitude & Longitude (e.g., WGS 84)
  - Projected: UTM, Lambert Conformal Conic
- Ensures accurate **measurement and overlay** of data.